EVERY PRO-ETALE COVERING OF AN OBJECT OF THE BIG PRO-ETALE SITE IS REFINED BY A COVERING OF THE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the covering by spectra of w-contractible rings, the refinement of each induced pro-etale covering by a finite disjoint open covering, and the collection of the pieces into a covering of the site.

Editorial note. The label lemma-proetale-induced names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-proetale-induced –start 6515 –end 6534 –local-id 28007 returned RESULT=OK (L28007 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Coverings by spectra of w-contractible rings. Same chapter, the lemma producing a covering of any object by affine objects which are spectra of w-contractible rings. Supplies the covering {V_k -> T} whose members are weakly contractible.
- Refinement of pro-etale coverings of a w-contractible affine scheme by finite disjoint open coverings. Same chapter, the lemma that every pro-etale covering of a w-contractible affine scheme is refined by a finite Zariski covering by a disjoint union. Supplies the refinement of each pulled back covering of V_k by a finite disjoint open covering.

Source reference. Original reference 098G: sources/stacks-proetale/pinned-original-source.tex, source lines 2465–2465. Prerequisite pointer invoked at proof line(s) 6518; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098I: sources/stacks-proetale/pinned-original-source.tex, source lines 3087–3087. Prerequisite pointer invoked at proof line(s) 6527; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 098F: sources/stacks-proetale/pinned-original-source.tex, source lines 2998–2998. Prerequisite pointer invoked at proof line(s) 6531; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Sch_{pro\text{-}étale}$ be a big pro-étale site as in Definition 098G. Let $T \in \mathrm{Ob}(Sch_{pro\text{-}étale})$. Let $\{T_i \to T\}_{i\in I}$ be an arbitrary pro-étale covering of T. There exists a covering $\{U_j \to T\}_{j\in J}$ of T in the site $Sch_{pro\text{-}étale}$ which refines $\{T_i \to T\}_{i\in I}$.*

Proof. Namely, we first let $\{V_k \to T\}$ be a covering as in Lemma 098I. Then the pro-étale coverings $\{T_i \times_T V_k \to V_k\}$ can be refined by a finite disjoint open covering $V_k = V_{k,1} \amalg \ldots \amalg V_{k,n_k}$, see Lemma 098F. Then $\{V_{k,i} \to T\}$ is a covering of $Sch_{pro\text{-}étale}$ which refines $\{T_i \to T\}_{i\in I}$. □